

UNIVERSITY OF KENT

MAST7865 MSc Data Science Project

Preparing the project report

1 Introduction

The project report is an important document for several reasons:

- It is used by the Examiners to assign a mark to your project. The project is marked by two University Examiners, one of whom is your Supervisor; the second Examiner has nothing other than the report from which to assess your project. Project reports are also seen by the External Examiner.
- The quality of the report itself is one of the assessment criteria.
- It is something concrete that you can take away from your MSc. For example, it can be shown to prospective employers.
- Many people have to write reports as part of their job. This is an opportunity to learn some of the skills of technical report writing.

These notes give some outline guidance on preparing your report.

2 How long will it take to write the report?

You probably need to spend something like 3 weeks writing the report. Many projects are spoiled by a rushed project report. You can help the writing process by keeping careful notes of what you are doing as you progress through the project. Also, there is no need to leave all the writing to the end. For example, if your project involves a complicated data set, you will have to understand the details of this fairly early in the project, so that may be a good time to write the part of the project report that describes the data.

Many people find it difficult to get started on writing. The general advice is just to get something down on paper (or on the computer screen). This might just be a list of the things you need to write about in a given section that can be expanded gradually. Don't worry about quality of writing at this stage, it's usually much easier to go back and edit text than it is to write it in the first place.

Make sure you allow plenty of time for editing. You will almost certainly need to edit several times to produce a good report. Producing a good layout of figures and tables can also be time consuming. In particular, you need to make sure that figures and tables are

numbered in some consistent and appropriate way and have proper captions to describe them.

3 Writing style

The writing style should be reasonably formal. This doesn't mean that you necessarily have to use long words or complicated sentence structures.

Clarity and precision are important. Ideally, someone who reads your project should gain enough information to understand exactly what you did in the project. For example, after reading a description of a simulation study, it should be possible for the reader to rerun the simulations if desired. This is the concept of *reproducible research* that is considered increasingly important.

A common problem in writing is to take for granted things that you may have discussed with your Supervisor, but which may well not be known to a more general reader (including the second Examiner).

In terms of background knowledge that you should assume, it's helpful to imagine that you are writing for a fellow MSc student. So, for example, you don't need to explain what a **t**-test is, but you do need to explain techniques that have not been covered in the main part of the course, or have been covered only briefly.

4 Use of English

We expect all projects to be written in clear English. Make sure you avoid spelling mistakes, which is generally easy to do in this era of spell-checkers. Try also to avoid grammatical mistakes, which detract from the professionalism of your report.

A previous External Examiner has suggested that students for whom English is not their first language might ask a native speaker to read through a draft of the project and advise on errors of grammar and syntax. This is acceptable *provided* advice is limited to issues of use of English and does not extend to the technical content of the work, which must of course be your own work (you will need to sign a declaration of this when you submit the project). In practice, this means asking someone who is not familiar with the statistical aspects at all, for example a student from another discipline.

5 Which software to use?

Any software is allowed as long as it produces a MS Word or PDF file. By this stage of the MSc you will already be quite experienced and should be able to find out how to do anything that you don't already know how to do.

6 Formatting

The project will be submitted electronically. It is up to you how you format it, but it is particularly important to make sure that subscripts, axis labels on figures, etc. are readable, so do not make things too small.

7 Structure of reports

A recommended format is as follows:

- **Title page**

This should include the project title, your name, the name of the supervisor and a statement such as ‘Submitted in partial fulfillment of the requirements of the degree of MSc Data Science, University of Kent, 2026’.

- **Summary**

A short summary of the project (maximum one page). You should emphasize the important conclusions here. This should be one of the last things you write. What would you want people to know about your project if they only had time to read this page?

- **Introduction**

Give background to the project and state in general terms what your objectives are. The end of this section should explain briefly (one or two sentences) what is contained in each of the remaining sections of the project. This is another section that is usually best completed after the majority of the report has been written.

- **Several sections making up the main project**

This is deliberately vague, because projects are very varied. For example, depending on your project, it may or may not be appropriate to have sections

- reviewing previous literature in detail (normally, the Introduction will contain some general references to earlier work);
- describing one or more data sets in detail;
- developing some new theory, if appropriate (many MSc projects do not develop any new theory);
- describing a computer program;
- describing results of some data analysis;
- describing the results of a simulation study

and so on.

- **Discussion**

This is where you can give an overview of what your project has achieved and the conclusions that you have reached. You should also discuss possibilities for further work – what would you do if you had more time to work on the project?

- **Acknowledgements**

Normally a single paragraph acknowledging any assistance you have had with the project. For example, you might wish to thank your supervisor, people who supplied any data that you worked on, anyone who helped you to get computer programs running, etc.

- **References**

These are important and are discussed separately below.

- **Appendices**

Examples of things that you might decide to include in Appendices are

- listings of data;
- listings of computer programs;
- additional simulation results.

This is only a recommended format – you don't have to follow it. For example, some people like to have a section involving numbered conclusions near the end (rather than incorporating conclusions as part of the Discussion). You might also like to include a table of contents.

8 References

References are important, but often badly done in projects. You need to do is choose and follow a referencing style. Your supervisor may have a recommendation.

<https://www.citethemrightonline.com/referencing-styles>

It doesn't matter which particular style you use, but it does matter that you use it correctly and consistently throughout. This means consistent style for referring to papers in the text and consistent style in the list of references near the end of the report. When your report is completed, you should go through and check that each reference in the text appears in the reference list and vice versa.

It is quite common nowadays to have references to material on the internet and this is sometimes unavoidable. In these cases, do give a "date accessed" along with a weblink. However, if you simply want to reference some standard statistical result, for example the properties of a particular distribution, it is better to refer to a conventional book or journal article rather than, say, Wikipedia, which may be less reliable.

9 Length of reports

There is a nominal recommended length of 50 pages (which does not count the appendices). However, this is only a very rough guide. Examiners will be assessing your project report against what might reasonably have been achieved within the project period. The quality of the work is important, not just the quantity.

10 What criteria are used to assess the project?

The criteria which the examiners consider when assessing the project are

- Depth – how deeply has the topic been studied, including study of the relevant literature.
- Quality of work – appropriateness of choice of techniques, accuracy, quality of computer work, appropriate use of methodology.
- Understanding – of the relevant theory, appropriateness of conclusions drawn, understanding of implications and limitations.
- Quality of writing – clarity of style, proper referencing of the literature, suitability of length of the project.
- Initiative/originality – to what extent have you been able to go beyond the suggestions of the supervisor and develop aspects of the project yourself?

The detailed criteria and module learning outcomes are available on the Moodle page.

11 Useful Links

<https://student.kent.ac.uk/studies/skills-for-academic-success>

<https://media.www.kent.ac.uk/se/52784/PlagiarismHowToAvoidIt.docx>

<https://www-citethemrightonline-com.chain.kent.ac.uk/>